On the Schrödinger Equation

Kieron Eisma

August 2026

Abstract

The Schrödinger Equation is an indispensable postulate of quantum mechanics that governs the evolution of the wave function:

$$i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r},t) + \hat{V}(\vec{r},t)\psi(\vec{r},t).$$

Because it is taken as a postulate in quantum theory, one cannot derive the Schrödinger Equation in the "same manner" as other equations in Physics, such as, for instance, the Euler-Lagrange Equation.

This begs the question of *why* the Schrödinger Equation was chosen as a postulate, instead of some other relation, and this is what the paper aims to answer.

Errors found in this paper, queries, along with suggestions are most welcome.

kiwi.eisma@gmail.com

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1 The Wave Packet

All wave functions in this paper are assumed to be Schwartz on the real line, and operators have not been explicitly indicated.

The wave equation is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u.$$

We shall consider the one-dimensional case for simplicity; the solutions to this equation yield plane waves that propagate along the x-axis. The general form of a wave packet is the superposition of these solutions:

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{i(kx-\omega t)}\, dk. \tag{1}$$

Define the initial wave function $\psi_0(x) = \psi(x,0)$, so that

$$\psi_0(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{ikx}\, dx,$$

where, by the Fourier Inversion Formula,

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi_0(x) e^{-ikx}\, dx.$$

The de Broglie Relation implies the momentum is given by $p = \hbar k$, where k is the wave vector ($k = \vec{k}$ in $\mathbb{R}^1$). It follows that $dk = \frac{dp}{\hbar}$. Moreover, the Planck Relation yields the energy: $E = \hbar\omega$. Therefore, Equation 1 becomes

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi\left(\frac{p}{\hbar}\right) e^{i\left(\frac{p}{\hbar}x - \frac{E}{\hbar}t\right)}\, \frac{dp}{\hbar}.$$

Now, define $\tilde{\phi}(p) = \frac{1}{\sqrt{\hbar}}\phi(k) = \frac{1}{\sqrt{\hbar}}\phi\left(\frac{p}{\hbar}\right)$. Then, we may express the wave packet above in the form

$$\begin{aligned}
\psi(x,t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{\hbar}\tilde{\phi}(p) e^{i\left(\frac{p}{\hbar}x - \frac{E}{\hbar}t\right)}\, \frac{dp}{\hbar} \\
&= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p) e^{\frac{i}{\hbar}(px - Et)}\, dp.
\end{aligned}$$

2 The Schrödinger Equation

In Section 1, we showed that the equation of a wave packet in $\mathbb{R}^1$ is given by the equation[1]

[1] Recall that the kinetic energy is given by $T = \frac{p^2}{2m}$.

$$\psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p) \exp\left[\frac{i}{\hbar}\left(px - \left(\frac{p^2}{2m} + V\right)t\right)\right] dp;$$

It follows that

$$\begin{aligned}
i\hbar\frac{\partial}{\partial t}\psi(x,t) &= \frac{1}{\sqrt{2\pi\hbar}}\left(i\hbar\frac{\partial}{\partial t}\int_{-\infty}^{\infty} \tilde{\phi}(p) \exp\left[\frac{i}{\hbar}\left(px - \left(\frac{p^2}{2m} + V\right)t\right)\right] dp\right) && (2)\\
&= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p)\left(\frac{p^2}{2m} + V\right) \exp\left[\frac{i}{\hbar}\left(px - \left(\frac{p^2}{2m} + V\right)t\right)\right] dp && (3)\\
&= \left(\frac{p^2}{2m} + V\right)\frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p) \exp\left[\frac{i}{\hbar}\left(px - \left(\frac{p^2}{2m} + V\right)t\right)\right] dp && (4)\\
&= \left(\frac{p^2}{2m} + V\right)\psi(x,t). && (5)
\end{aligned}$$

However, recall that the momentum operator in position space is given by[2]

$$\vec{p} = -i\hbar\vec{\nabla},$$

which is to say $p^2 = \hbar^2\nabla^2$, so in $\mathbb{R}^1$,

$$p^2 = \hbar^2\frac{\partial^2}{\partial x^2}.$$

Substituting this into Equation 5,

$$\begin{aligned}
i\hbar\frac{\partial}{\partial t}\psi(x,t) &= \left(\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V\right)\psi(x,t)\\
&= \hat{H}\psi(x,t),
\end{aligned}$$

where $\hat{H}$ denotes the Hamiltonian. This is the (time-dependent) one-dimensional Schrödinger Equation.

References

[Zet22] Nouredine Zettili. *Quantum Mechanics*. Wiley, 2022. ISBN: 978-1-118-30789-2.

[2]See Equation 2.346 in [Zet22]